

Comparative Test of 3D-PAWS Anemometers Against a Pitot Tube Reference

Wind tunnel: brief performance analysis



Figure 1. Wind tunnel used for the tests.

Sensor description. The tested anemometers follow the 3D-PAWS project design and were manufactured by 3D printing. Rotation is detected with a Honeywell SS451A Hall effect sensor and two magnets. In the intended configuration, the passage of both magnets registers two pulses per revolution. All speeds reported in this document are in meters per second (m/s).

Background on anemometer 087. During inspection of the unit, one of the two magnets was found to be misplaced, so its original reading was approximately half the expected value. For this reason, the analysis focuses on the anem087 $\times 2$ series, used as an approximation of the response that would be obtained with two pulses correctly registered per revolution.

Combined analysis criterion. In this version, all available tests are analyzed as a single experimental set, without separating them by test date. Each series is compared against the Pitot tube speed recorded in the same test. For anemometers 001, 012, 014, and 015, the average of the available readings at each reference point is used.

1. Objective

To jointly compare the performance of anemometers 001, 012, 013, 014, 015, and 087 × 2 against the wind tunnel reference (Pitot tube), regardless of when each test was performed; to quantify linearity, bias, and relative error; to assess on an exploratory basis whether the different material and surface finish of anemometer 015 may be associated with a different response; and to verify whether the acquisition system used with anem013 (Raspberry Pi or Particle) introduces appreciable differences.

2. Combined comparison with the reference

The results were integrated into a single comparison. In tests 001, 012, 014, and 015, each point is the average of approximately 9 to 15 readings. For anemometers 013 and 087 × 2, repeated points from the ascending and descending sweeps were averaged where applicable. The 5 to 30 m/s range is used as the main reference for performance indicators, since a different start-up regime appears around 1 m/s.

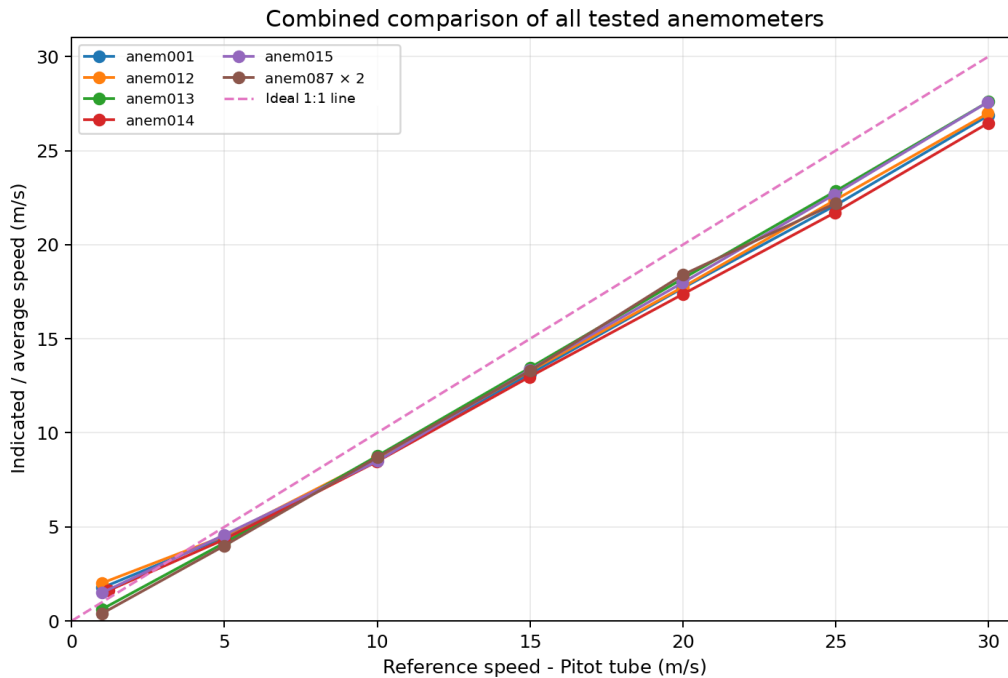


Figure 2. Combined comparison of all anemometers against the speed measured with the Pitot tube.

Anemometer	Material	Points 5-30	Slope	R ²	Bias (m/s)	Mean rel. error
anem001	ASA Polymaker	6	0.897	0.9997	-2.02	-11.8%
anem012	ASA Polymaker	6	0.903	0.9997	-1.90	-11.0%
anem013	ASA Polymaker	6	0.938	1.0000	-1.66	-10.9%
anem014	ASA Polymaker	6	0.885	0.9997	-2.27	-13.3%
anem015	ASA Elegoo	6	0.928	0.9991	-1.71	-10.3%
anem087 × 2	ASA Polymaker	5	0.922	0.9984	-1.68	-12.7%

Between approximately 5 and 30 m/s, all units show a strongly linear response ($R^2 \geq 0.998$) and a negative bias relative to the Pitot. Mean relative error ranges from -10.3% to -13.3%. Anemometer 015 has the lowest mean relative error of the set (-10.3%), while anemometer 013 has a slightly

smaller mean bias in absolute terms (-1.66 m/s versus -1.71 m/s for 015).

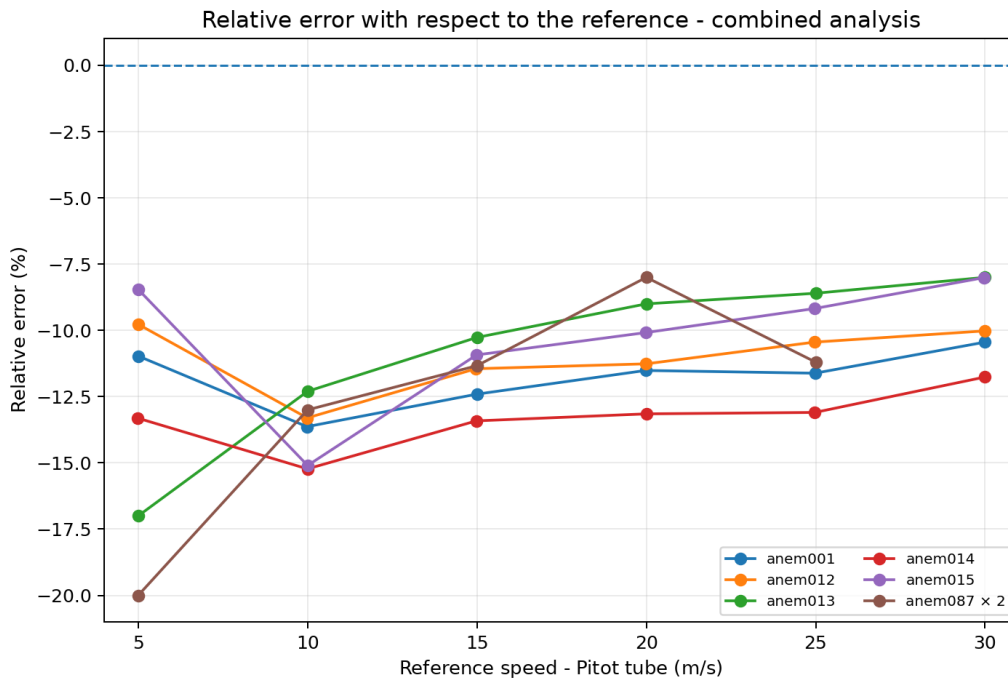


Figure 3. Relative error of all anemometers with respect to the reference in the main test range.

Dispersion between units is greater in the range near 1 m/s. Anemometers 001, 012, and 015 read above the Pitot on average; 013 and 087 × 2 read below; and 014 recorded no reading at 1.0 m/s, although it responded when the reference was raised to 1.2 m/s. For this reason, the start-up threshold should be analyzed separately from the quasi-linear response observed at higher speeds.

3. Influence of the acquisition board

Measurements of anem013 with Raspberry Pi and Particle are practically equivalent at the four comparable points. The mean absolute difference is 0.055 m/s and the maximum observed is 0.10 m/s. With the available data, the acquisition board has no apparent relevant influence on the calculated speed.

Reference (m/s)	Raspberry Pi (m/s)	Particle (m/s)
10	8.74	8.70
15	13.32	13.40
20	18.20	18.20
25	22.80	22.90

4. Printing material and surface finish

Anemometers 001, 012, 013, 014, and 087 were printed in Polymaker ASA. Anemometer 015 was printed in Elegoo ASA and, based on the team's visual observation, has a smoother surface finish. This difference is included as a potentially relevant manufacturing variable, although it was not measured instrumentally.

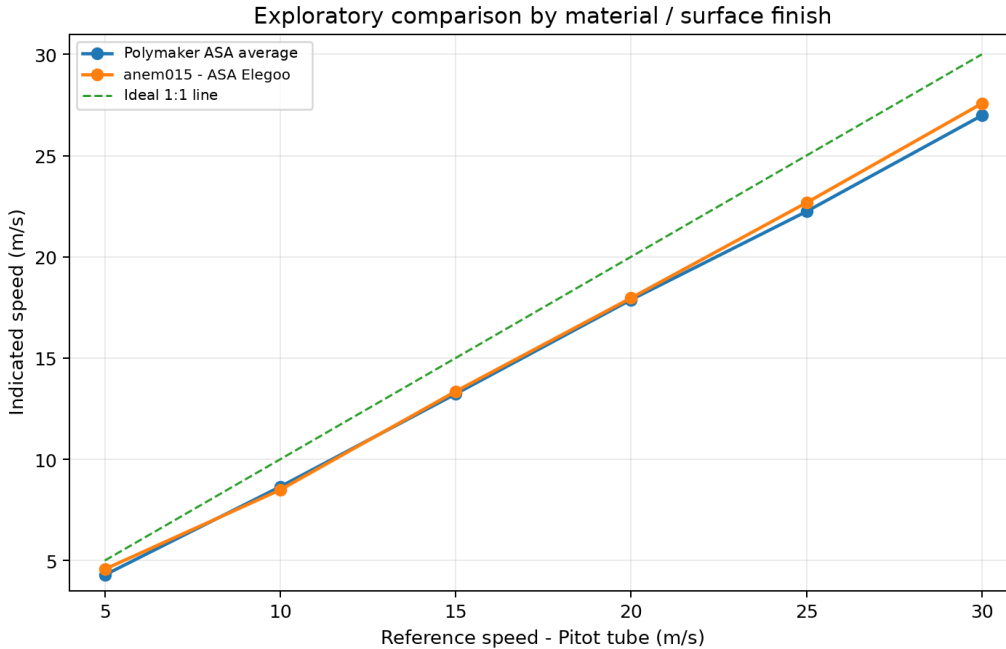


Figure 4. Exploratory comparison between anemometer 015 (Elegoo ASA) and the average of the units printed in Polymaker ASA.

Pitot (m/s)	Polymaker avg.	anem015 Elegoo	Diff. 015-PM (m/s)	Diff. (%)	n PM
5	4.287	4.573	+0.286	+6.7%	5
10	8.645	8.491	-0.154	-1.8%	5
15	13.226	13.353	+0.127	+1.0%	5
20	17.879	17.966	+0.087	+0.5%	5
25	22.238	22.670	+0.432	+1.9%	5
30	26.979	27.571	+0.593	+2.2%	4

At the nominal points from 5 to 30 m/s, anemometer 015 reads on average +0.23 m/s relative to the average of the Polymaker units, approximately +1.7%. The difference is not uniform: at 10 m/s, 015 is slightly below the Polymaker average, while at 25 and 30 m/s it is about 0.43 and 0.59 m/s above, respectively. At 30 m/s, the Polymaker average is calculated with four units because 087 has no point there.

The smoother finish of 015 could, as a hypothesis, change the aerodynamic drag of the exposed surfaces and/or the mechanical behavior of the rotor, producing a response somewhat closer to the reference. However, these tests do not allow causality to be attributed to the material or roughness: there is only one Elegoo unit, the tests were performed at different times, and printing tolerances, assembly, bearings, alignment, magnets, and other unit-to-unit variations may play a role. The result should therefore be interpreted as a signal to verify, not as proof that Elegoo ASA improves

performance.

To test this effect specifically, it would be advisable to print several geometrically equivalent units in Polymaker ASA and Elegoo ASA using the same printer, orientation, layer height, and print parameters; test them alternately under the same conditions; and, if possible, quantify surface roughness. That design would make it possible to separate the effect of material/finish from the inherent variability of each anemometer.

Data note: in the original 015 test file, the fifth point was listed as 14.981 m/s. This report uses 19.981 m/s, a value later confirmed for that test.

5. Scope and conclusions

- The combined analysis of anemometers 001, 012, 013, 014, 015, and 087 × 2 shows a strongly linear response against the Pitot tube between approximately 5 and 30 m/s, even though the tests were performed at different times.
- In that range, all units tend to underestimate the reference. Mean relative error is approximately -10% to -13%, so an individual correction or calibration would improve agreement with the reference.
- Anemometer 015, printed in Elegoo ASA with a visually smoother finish, has the lowest mean relative error of the set (-10.3%) and reads, on average, about 0.23 m/s above the Polymaker units between 5 and 30 m/s. The difference is small and does not by itself prove a material effect.
- A possible influence of roughness/surface finish is a reasonable hypothesis from an aerodynamic and mechanical standpoint, but it requires controlled tests with replicas manufactured under equivalent conditions in order to isolate it from unit-to-unit variability.
- The zone near 1 m/s shows greater dispersion and non-linear start-up behavior. Anemometer 014 recorded no signal at 1.0 m/s and began responding when the reference was raised to 1.2 m/s.
- No relevant differences were detected between Raspberry Pi and Particle as acquisition systems for anemometer 013 at the four comparable points.
- These results constitute an experimental characterization. A formal metrological calibration would need to incorporate the traceability and uncertainty of the reference, the environmental conditions of each test, and a dedicated repeatability and reproducibility campaign.



Figure 5. Anemometer mounted during the test.

Annex. Test measurements

All speeds are in m/s.

A. Anemometer 013 with Raspberry Pi

Pitot (m/s)	anem013 (m/s)
1	0.66
5	4.15
10	8.74
15	13.32
20	18.20
25	22.80
30	27.60
25	22.90
20	18.20
15	13.60
10	8.80
5	4.15
1	0.60

B. Anemometer 087 × 2

Pitot (m/s)	anem087 × 2 (m/s)
1	0.40
5	4.00
10	8.60
15	13.40
20	18.40
25	22.20
20	18.40
15	13.20
10	8.80
5	4.00
1	0.40

C. Anemometer 013: board comparison

Pitot (m/s)	Raspberry Pi (m/s)	Particle (m/s)
10	8.74	8.70
15	13.32	13.40
20	18.20	18.20
25	22.80	22.90

D. Anemometer 001

Pitot (m/s)	n	Mean (m/s)	Std. dev. (m/s)
0.996	9	1.762	0.816
5.004	11	4.455	0.185
9.983	10	8.622	0.120
15.001	11	13.140	0.103
19.977	10	17.677	0.130
24.982	12	22.080	0.142
29.996	12	26.863	0.148

E. Anemometer 012

Pitot (m/s)	n	Mean (m/s)	Std. dev. (m/s)
0.989	12	2.009	0.615
4.996	10	4.508	0.166
9.990	11	8.661	0.071
14.973	11	13.259	0.164
19.995	10	17.743	0.068
24.960	12	22.353	0.118
29.986	10	26.981	0.097

F. Anemometer 014

Pitot (m/s)	n	Mean (m/s)	Std. dev. (m/s)
1.000	0	No reading	-
1.200	13	1.613	0.383
4.988	12	4.324	0.185
9.996	15	8.474	0.146
14.983	10	12.973	0.124
20.007	10	17.376	0.141
24.976	11	21.705	0.066
29.998	10	26.470	0.175

G. Anemometer 015

Pitot (m/s)	n	Mean (m/s)	Std. dev. (m/s)
1.001	10	1.507	0.702
4.995	10	4.573	0.157
10.001	10	8.491	0.579
14.991	10	13.353	0.097
19.981	10	17.966	0.074
24.961	10	22.670	0.124
29.972	10	27.571	0.110

Data sources: Comparaciones 12ago.xlsx and test files 001, 012, 014, and 015. For the analysis, all series were integrated against their contemporaneous Pitot reference. For anemometer 015, 19.981 m/s is used for the fifth point, per later confirmation. Material as reported by the team: Elegoo ASA for 015 and Polymaker ASA for the remaining units; the greater smoothness of 015 is a visual observation. Photographs provided by the team.

ANNEX I: CALIBRATION

Comparison of original and corrected results of the 3D-PAWS anemometers against the Pitot tube reference

Calibration criterion

The linear calibrations were obtained using the main test range between 5 and 30 m/s. The zone near 1 m/s is kept in the charts for illustrative purposes, but was not used to calculate the coefficients due to the start-up behavior and greater dispersion observed at low speeds.

1. Uncalibrated results

In their original condition, all anemometers show an approximately linear response, but with a systematic underestimation of speed relative to the Pitot tube in the main test range.

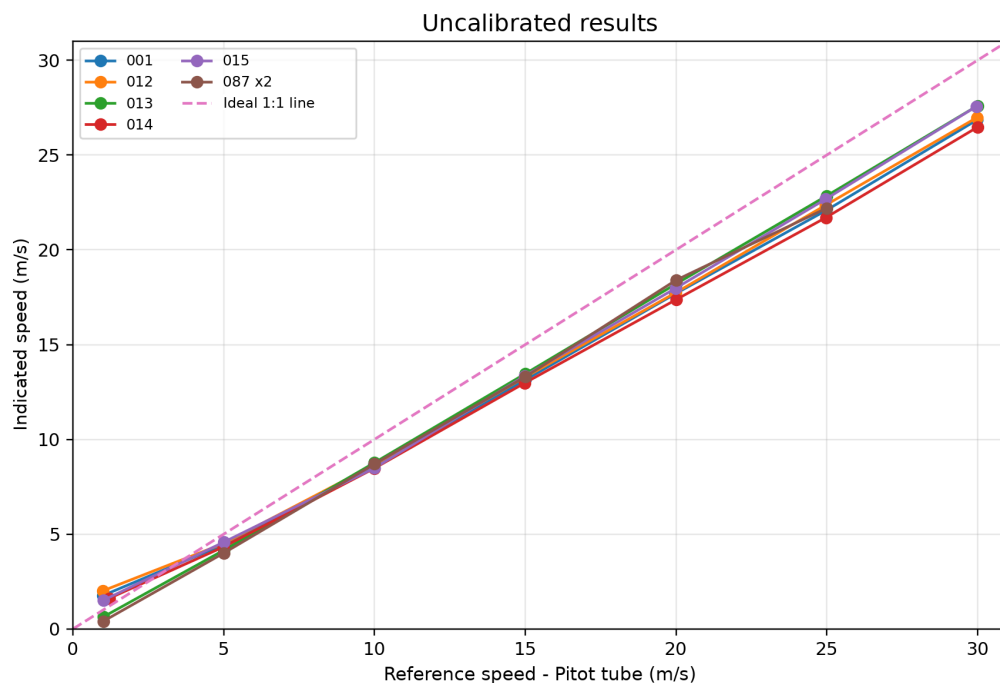


Figure 1. Original response of all anemometers against the reference.

2. Individual linear calibration

For each anemometer, an independent linear equation of the form $V_{meas} = a \cdot V_{pitot} + b$ was fitted. The correction applied is $V_{corr} = (V_{meas} - b) / a$.

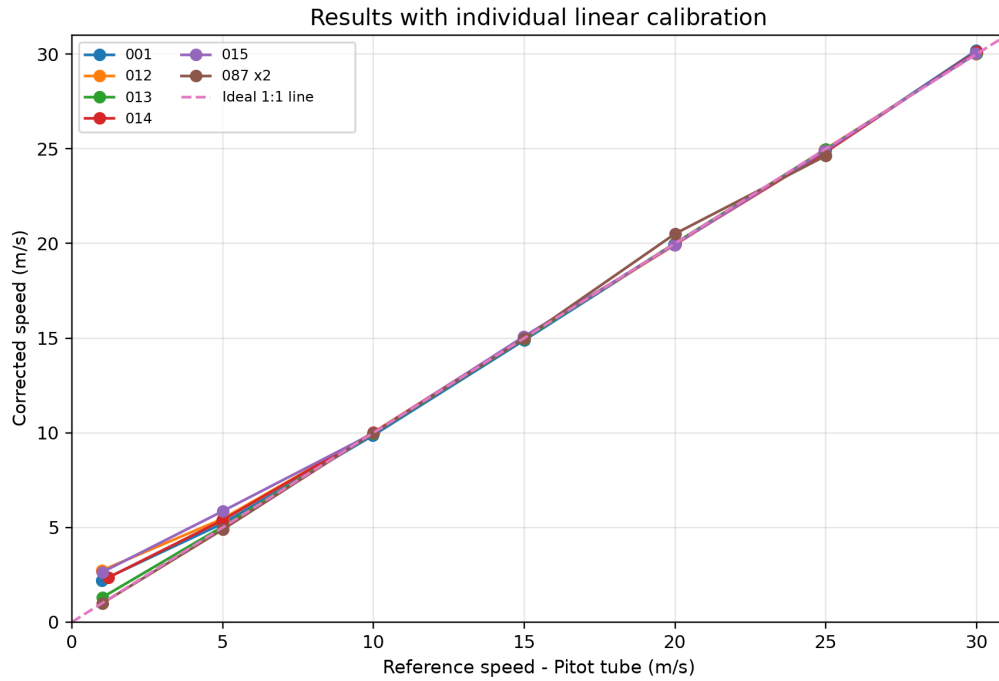


Figure 2. Results after applying an individual linear calibration.

Anem.	a	b (m/s)	R ²	Correction equation
001	0.8972	-0.220	0.9997	$V_{corr} = (V_{meas} + 0.220) / 0.8972$
012	0.9151	-0.484	1.0000	$V_{corr} = (V_{meas} + 0.484) / 0.9151$
013	0.9385	-0.585	1.0000	$V_{corr} = (V_{meas} + 0.585) / 0.9385$
014	0.8945	-0.484	0.9998	$V_{corr} = (V_{meas} + 0.484) / 0.8945$
015	0.9512	-0.996	0.9999	$V_{corr} = (V_{meas} + 0.996) / 0.9512$
087 x2	0.9220	-0.510	0.9984	$V_{corr} = (V_{meas} + 0.510) / 0.9220$

3. General linear calibration

A single line was also obtained using all points from all anemometers between 5 and 30 m/s combined. This alternative allows the same correction to be applied to all units.

$$\begin{aligned}\text{General model: } V_{\text{meas}} &= 0.91700 \cdot V_{\text{pitot}} - 0.48961 \\ \text{General correction: } V_{\text{corr}} &= (V_{\text{meas}} + 0.48961) / 0.91700 \\ \text{Global } R^2 &: 0.9982\end{aligned}$$

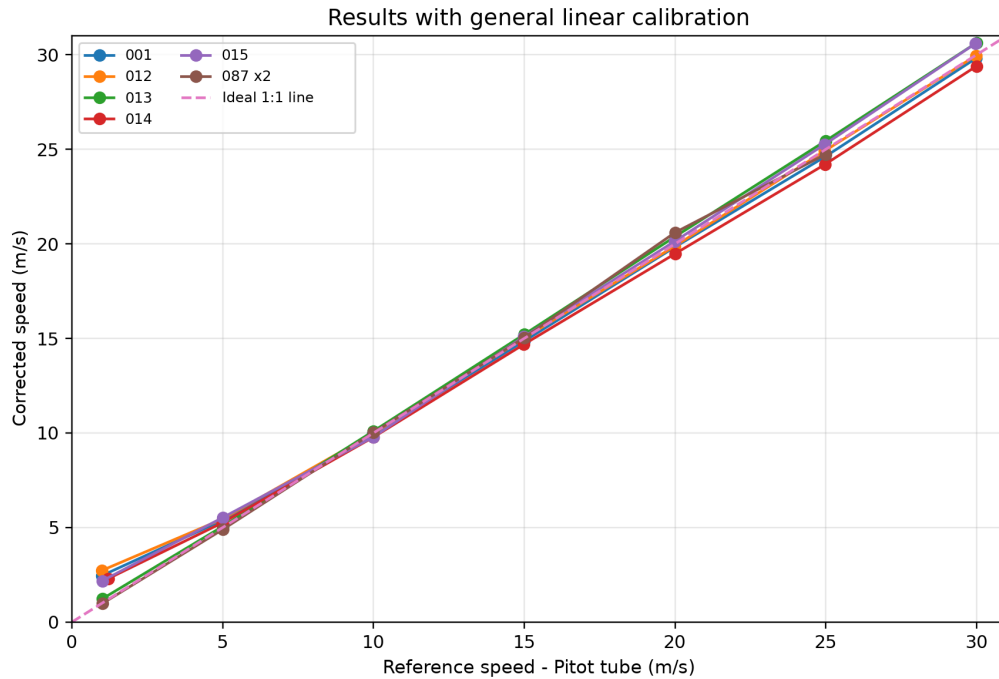


Figure 3. Results after applying a single general linear calibration.

4. Summary comparison

The following table shows the mean absolute error relative to the Pitot, calculated only in the 5-30 m/s range. The individual calibration minimizes the error of each unit; the general calibration maintains a significant improvement with a single common equation.

Anem.	Uncalibrated (m/s)	Individual calib. (m/s)	General calib. (m/s)
001	2.018	0.132	0.213
012	2.181	0.031	0.044
013	1.662	0.032	0.306
014	2.592	0.085	0.484
015	1.971	0.062	0.276
087 x2	1.680	0.204	0.204

Conclusion: the results show that a linear calibration adequately corrects the underestimation observed between 5 and 30 m/s. The individual calibration provides the best fit per unit, while the general calibration is a simple alternative for applying a single equation to the whole set.